BUSINESS INTELLIGENCE OVERVIEW

AnalystLab Africa | Week 2 | Superstore Sales Dataset

1. Introduction to Business Intelligence

Business Intelligence (BI) refers to the processes, technologies and practices used to collect, organise, analyse and present business data in a form that supports better decision-making. It helps an organisation move beyond storing information to understanding what its data reveals about performance, customers, products and operations.

2. Why organisations use dashboards

A dashboard brings important measures and visual summaries together in one place. Instead of reviewing multiple spreadsheets or lengthy reports, managers can monitor key performance indicators (KPIs), compare results and identify patterns more efficiently. Dashboards present information through charts, tables, maps and interactive filters, making it easier to explore performance.

3. How dashboards improve business decisions

- Monitor performance against selected measures and identify changes over time.
- Compare regions, customer groups and product categories to understand where results are strongest.
- Identify exceptions, such as products or categories generating low or negative profit.
- Investigate performance through interactive filters rather than relying only on static summaries.
- Use evidence to prioritise further analysis, operational changes and resource allocation.

4. Business objective of the Superstore analysis

The objective of this case study is to analyse the Superstore sales dataset using Microsoft Power BI and develop an interactive dashboard that provides an overview of sales and profitability performance. The analysis considers total sales, total profit, number of orders, average order value and profit margin. It also examines performance by region, product category, sub-category, customer segment, state and time period.

5. Expected value of the analysis

By combining summary KPIs with detailed visual analysis, the dashboard can help identify where revenue is generated, where profit is strongest or weakest, and which areas warrant further investigation. The interactive slicers allow users to explore results by year, region, category and customer segment. The resulting insights can support more informed decisions about product performance, market focus and profitability improvement.

Part 2: DATA PREPARATION

1. Dataset overview

The Superstore dataset was imported into Microsoft Power BI for preparation and analysis. It contains 9,994 transaction-line records across 21 columns and covers 2014 to 2017. The fields include order and shipping dates, customer details, geographical information, product categories and sub-categories, sales, quantity, discount and profit.

2. Data type review and preparation

Data types were reviewed in Power Query and corrected to support accurate analysis. Order Date and Ship Date were converted to Date type. Sales, Profit and Discount were set to decimal number, while Quantity and Row ID were set to whole number. Identifier and descriptive fields, including Order ID, Customer ID, Product ID, names, categories, regions and postal codes, were treated as text where appropriate.

3. Data quality checks

The preparation included checking missing values, blank rows and blank columns. All 21 columns contained data. The exact-duplicate check across all 21 columns identified zero exact duplicate rows. The Remove Blank Rows function was applied as part of the blank-row check.

4. Business-rule validation

The review identified 1,871 transaction-line records with negative profit, approximately 18.72% of the 9,994 records. No negative or zero sales values were identified. No discount values below zero or above one were identified. A date validation check found no records where Ship Date preceded Order Date.

5. Treatment of negative-profit records

Negative-profit records were retained rather than automatically removed. They may represent genuine business outcomes where discounts, pricing or other transaction economics result in a loss. Retaining them allows the dashboard to identify product areas where profitability may require further investigation.

6. Preparation outcome and limitations

The preparation process established consistent data types and included checks for duplicate rows, missing values, blank rows and columns, unusual sales and discount values, negative-profit records and the logical order of order and shipping dates. The prepared dataset was then used for KPI creation and dashboard development. This report records the checks and results confirmed during the working process; it does not claim that every possible data-quality issue or every column-level error scenario has been independently audited.

Part 4: BUSINESS INSIGHTS

1. Identify 5 Key Business Insights

Area	Finding
Insight 1	Technology leads both sales ($836,154.03) and profit ($145,454.95), with a 17.40% margin.
Insight 2	Furniture generates $741,999.80 in sales but has a low 2.49% profit margin.
Insight 3	Office Supplies has the lowest category sales ($719,047.03), but a healthy 17.04% margin.
Insight 4	Sub-category profitability varies considerably: Copiers have a 37.19% margin, while Machines have only 1.79%.
Insight 5	Monthly sales fluctuate, with increases followed by dips and recoveries rather than a consistent upward trend.

2. Identify 3 Business Risks

1. Low profitability in Furniture: Despite generating substantial sales, Furniture has a profit margin of only 2.49%. This creates a risk that costs, discounts or pricing decisions may be reducing the value of its sales.

2. Loss-making sub-categories: Tables, Bookcases and Supplies recorded negative profit margins of -8.56%, -3.02% and -2.55%, respectively. Continuing to sell these products without reviewing their costs and pricing could continue to affect overall profitability.

3. Uneven sales distribution: More than half of total sales come from the Consumer segment, while sales also vary considerably across states and regions. This may expose the business to changes in demand in its strongest markets. The data suggests concentration, although it does not establish that the business is dependent on those markets.

3. Identify 3 Business Opportunities

1. Expand high-margin product performance: Copiers and Accessories have relatively strong profit margins. Management could investigate opportunities to promote these products, improve their availability and increase sales where customer demand supports it.

2. Improve Furniture profitability: Review the pricing, discounting, supplier costs and operating costs associated with Tables, Bookcases and other weak-performing Furniture products. The aim should be to improve margins rather than simply increase sales volume.

3. Develop Corporate and Home Office customer segments: Consumer customers account for 50.56% of sales. The business could explore targeted marketing and customer-retention strategies to grow Corporate and Home Office sales while maintaining its Consumer customer base.

Part 5: BUSINESS RECOMMENDATIONS

Recommendation 1 - Improve Furniture profitability

Finding: Furniture generates substantial sales but has a profit margin of only 2.49%.

Recommended action: Management should conduct a product-level profitability review of the Furniture category, particularly Tables and Bookcases. This should include reviewing supplier costs, selling prices, discount levels and operating costs.

Expected outcome: Improved profit margins and reduced losses from underperforming products.

Recommendation 2 - Review loss-making sub-categories

Finding: Tables, Bookcases and Supplies have negative profit margins.

Recommended action: Investigate the reasons for their losses. Management should consider adjusting prices, renegotiating supplier arrangements, reducing excessive discounts or reconsidering products that remain unprofitable.

Expected outcome: Reduced losses and better control over product-level profitability.

Recommendation 3 - Prioritize high-margin products

Finding: Copiers and Accessories have comparatively strong profit margins.

Recommended action: Explore targeted promotions, improved stock availability and sales strategies for these products. Management should monitor demand and inventory before increasing stock commitments.

Expected outcome: Growth in revenue from products that contribute stronger profit margins.

Recommendation 4 - Diversify customer segment performance

Finding: Consumer customers contribute approximately 50.56% of total sales.

Recommended action: Develop targeted marketing and customer-retention initiatives for Corporate and Home Office customers. For example, the business could explore business-customer packages, repeat-purchase incentives and tailored product offers.

Expected outcome: Stronger sales across customer segments and reduced reliance on one segment.

Recommendation 5 - Improve monthly sales planning

Finding: Monthly sales fluctuate rather than showing a consistent upward trend.

Recommended action: Management should compare monthly performance across the four years to identify recurring peaks and weaker periods. Use those findings to plan promotions, inventory levels, staffing and sales targets.

Expected outcome: Better demand planning, more effective resource allocation and improved readiness for periods of weaker sales.